

SUBODH WAGH

Ahilyanagar, Maharashtra | +91 8830278326

subodhwagh1122@gmail.com | [linkedin.com/in/subodh-wagh](https://www.linkedin.com/in/subodh-wagh) | github.com/subodh2003

Professional Summary

M.Sc. Computer Science student (Graduating 2026) focusing directly on Linux networking topologies, low-level systems execution, and custom infra automation. Handbuilt core utilities including custom concurrent Go engines, isolated network fabrics, and parser backends from scratch to understand OS-level constraints.

Technical Skills

Languages: C, Python, Go, Bash scripting, x86 Assembly

Networking: BGP routing, EVPN (Type-2/5 mechanics), VXLAN tunneling, Symmetric IRB routing, VRFs, raw TCP/IP sockets

Linux Internals: Network Namespaces, Kernel-level isolation, Linux Bridges, Veth pairing, Multithreading execution

Tools & Infrastructure: FRRouting (FRR system), Vtysh engine, NeST Emulator, Docker setups, Kubernetes networking, tcpdump, Wireshark parsing, Git versioning

Academic & Independent Projects

High-Performance Remote KV Cache Vault | [\[View Code\]](#)

2026

Go, Linux TCP Sockets, Systems Programming, Concurrency

- Wrote a multithreaded network storage vault in Go using custom framing layouts over bare TCP sockets to map quick memory swappings for GPU contexts.
- Fixed lock contention issues by rewriting the data layer into a 256-shard segmented map layout, avoiding heavy global system pauses during peak threads.
- Eliminated interleaving deadlocks via strict internal index rank checks, keeping operations clean during brutal multi-key write stress routines.

EVPN-VXLAN Data Center Fabric Emulator | [\[View Code\]](#)

2025 – 2026

Python, NeST Network Emulator, Linux Namespaces, FRRouting, BGP, VXLAN

- Built a functional Python automation layer using the NeST API to spin up, configure, and teardown multi-tier Spine-Leaf data center topologies instantly.
- Isolated routing domains manually inside bare Linux by tying together network namespaces, virtual wires (veth), and dedicated tenant VRF profiles.
- Set up a programmatic control plane to dynamically push configurations down to FRRouting daemons for underlay BGP peering and Symmetric L3 overlays.

Custom Recursive-Descent Compiler | [\[View Code\]](#)

2025

C, x86-64 Assembly

- Handbuilt a lexical analyzer and recursive-descent parser in pure C to process custom syntax rules straight down into raw x86-64 instructions.
- Managed memory scopes and variable lifecycles using custom symbol tables and strict manual register allocation tracking.
- Handled state translations for conditional branches, loops, and stack frames while preserving execution semantics.

Two-Pass x86 Instruction Assembler | [\[View Code\]](#)

2025

Python, x86 ISA

- Implemented a two-pass assembler architecture in Python to handle forward label references and compute instruction relative offsets.
- Coded bitwise translation logic to manually emit machine code, managing layout tasks like ModR/M byte evaluations and verifying outputs against NASM.

Education

M.Sc. Computer Science – Savitribai Phule Pune University

2024 – 2026

CGPA: 7.82 / 10 — Studies: Distributed Systems, Compiler Design, Operating Systems, Computer Networks

B.Sc. Computer Science – PVP College, Ahilyanagar

2021 – 2024

CGPA: 8.6 / 10

Leadership

- **Core Member, CSFG Cell (SPPU):** Organized technical seminars and departmental initiatives.